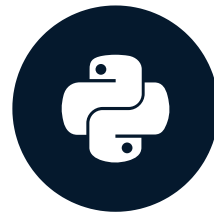


Introduction to Python for Finance

INTRODUCTION TO PYTHON FOR FINANCE

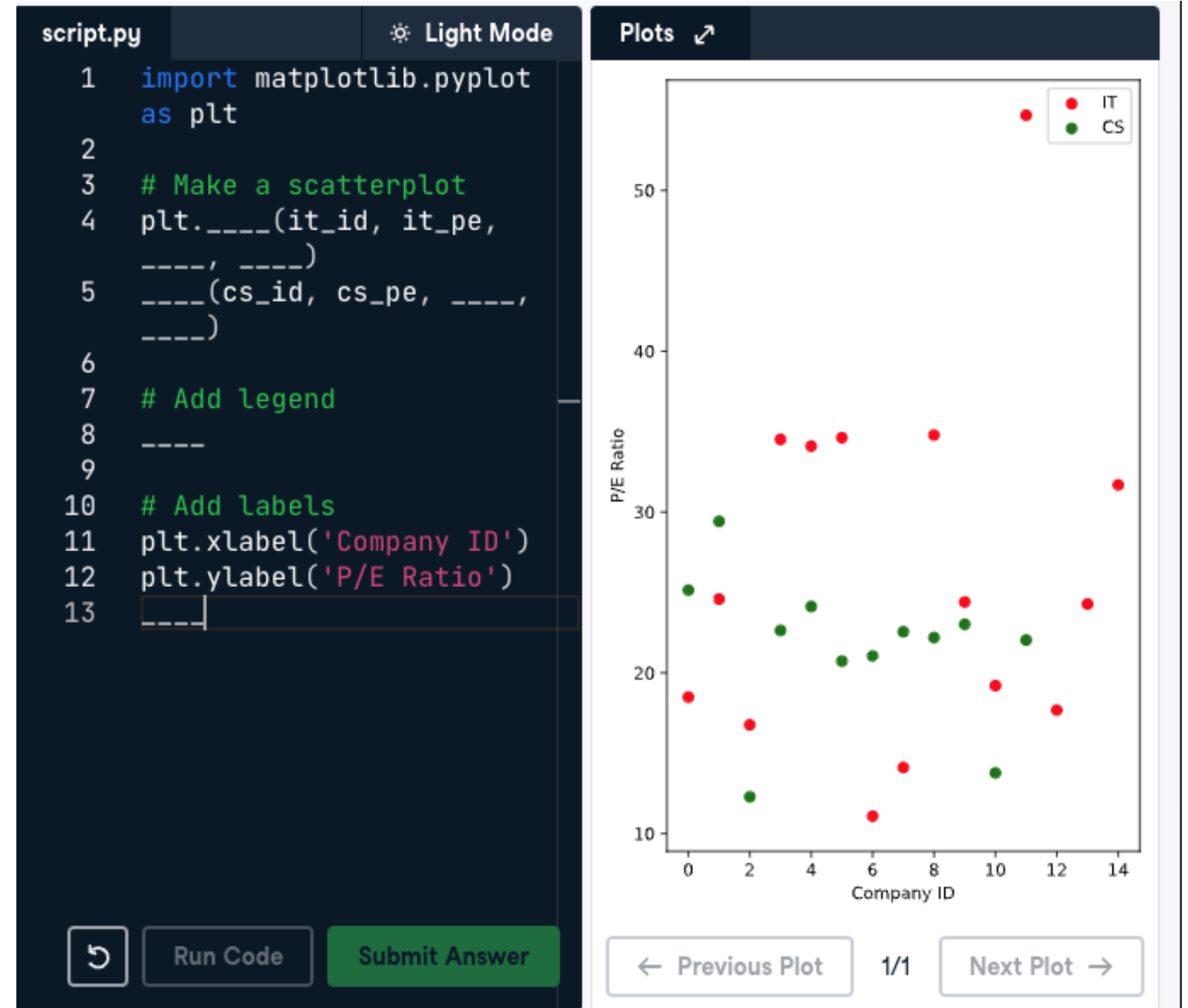
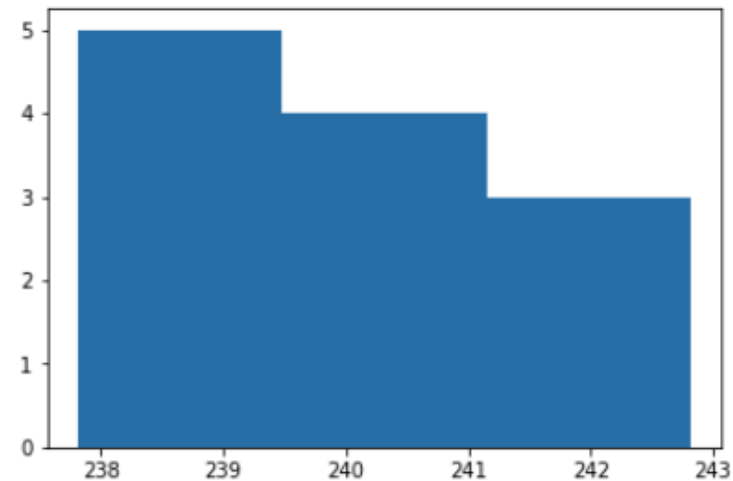


Adina Howe
Associate Professor

Coming up...

Histograms and matplotlib.pyplot

```
import matplotlib.pyplot as plt
plt.hist(x=prices, bins=3)
plt.show()
```



Why Python for Finance?

- Easy to Learn and Flexible
 - General purpose
 - Dynamic
 - High-level language
- Integrates with other languages
- Open source
 - Accessible to anyone





←

≡ Course Outline

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Exercise

<

DataCamp's exercise introduction.

Instructions

100 XP

DataCamp's exercise instructions.

💡 Take Hint (-30 XP)

script.py

Light Mode

1

Commands here will be saved as a Python Script.

↺

Run Code

Submit Answer

IPython Shell

▼

In [1]: 1 + 1

Out[1]: 2

In [2]: 2 ** 3

Out[2]: 8

In [3]:

The IPython Shell - commands here can be executed interactively.

Common mathematical operators

Operator	Meaning
+	Add
-	Subtract
*	Multiply
/	Divide
%	Modulus (remainder of division)
**	Exponent

Common mathematical operators

```
# Print the sum of 8 and 4  
print(8 + 4)
```

12

```
# Print 8 divided by 4  
print(8 / 4)
```

2

- Explicitly use `print()`

Any comments?

```
# Print 8 divided by 2  
print(8 / 2)
```

```
# Print 2 raised to the power of 2  
print(2**2)
```

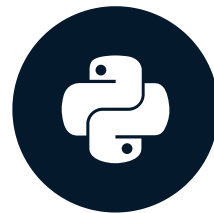
```
# Print 1.0 add 0.10  
print(1.0 + 0.10)
```


Let's practice!

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Variables

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Associate Professor

Define variables

Store information that can be referenced in code

- **Examples:** `msft_stock` , `interest_rate` , `bonds`
- Name = Value: `interest_rate = 0.05`

Variable names

- Upper or lower case letters, digits, and underscores
- *Cannot* start with a digit
- Some variable names are *reserved* in Python (e.g., `class` or `type`) and should be avoided

Defining variables in Python

```
# Correct
```

```
day_2 = 5
```

```
# Incorrect, variable name starts with a digit
```

```
2_day = 5
```

Using variables to evaluate stock trends

$$\text{Price to earning ratio} = \frac{\text{Market price}}{\text{Earnings per share}}$$

```
price = 200
earnings = 5
pe_ratio = price / earnings
print(pe_ratio)
```

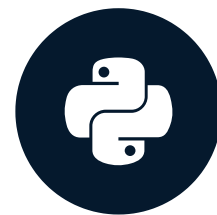
40

Let's practice!

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Variable Data Types

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Adina Howe
Instructor

Python Data Types

Variable Types	Example
Strings	'hello world'
Integers	40
Floats	3.1417
Booleans	True or False

Variable Types

Variable Types	Example	Abbreviations
Strings	'Tuesday'	<code>str</code>
Integers	40	<code>int</code>
Floats	3.1417	<code>float</code>
Booleans	True or False	<code>bool</code>

What data type is a variable: `type()`

To identify the type, we can use the function `type()` :

```
type(variable_name)
```

```
pe_ratio = 40  
print(type(pe_ratio))
```

```
<class 'int'>
```

Variable manipulations

```
x = 5  
print(x * 3)
```

15

```
print(x + 3)
```

8

```
y = 'stock'  
print(y * 3)
```

'stockstockstock'

```
print(y + 3)
```

TypeError: must be str, not int

Changing variable types

```
pi = 3.14159  
print(type(pi))
```

```
<class 'float'>
```

```
pi_string = str(pi)  
print(type(pi_string))
```

```
<class 'str'>
```

```
print('I love to eat ' + pi_string + '!!')
```

```
I love to eat 3.14159!
```

Let's practice!

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